

**UNIVERSITI TEKNOLOGI MALAYSIA
FACULTY OF COMPUTING**

**TEST 2
SEMESTER I 2023/2024**

SUBJECT CODE : SECJ2013
SUBJECT NAME : DATA STRUCTURES AND ALGORITHMS
YEAR/COURSE : 2 (SECJH/ SECVH/ SECBH/ SECRH/ SECPh)
TIME : 8.00 PM – 10.00 PM
DATE : 3rd JANUARY 2024 (WEDNESDAY)
VENUE : N28 MPK1-MPK10, CGMTL, CCNA, CSL, MCASE, MJIT

INSTRUCTIONS TO THE STUDENTS:

- This is a **CLOSED-BOOK** practical test. The test will require you to code on a computer.
- References to any resources, including any source code, websites, etc., are strictly prohibited.
- You should code **OFFLINE** using any C++ IDE such as VS Code, DevCPP, etc.
- The use of any form of artificial intelligence (AI) tool is strictly prohibited. Such tools include ChatGPT, AI-based VS Code extensions like Co-Pilot, etc.
- Be warned that your program will undergo a screening process for plagiarism and AI detection.
- You must use only the computers in the lab. You are not allowed to use your own laptop to attempt this test.
- All comments in the program will not be graded.
- Programs that cannot compile will receive a penalty.
- Any question related to the test question will not be entertained at all. Please read the question carefully.

IMPORTANT NOTES:

- **Starter** codes are provided for this test.
- You need to write the program **in the starter code**.
- Write each program using **inline style** and use only a **single source code file**.

This question paper consists of FIVE (5) printed pages including this page.

Question 1

[50 Marks]

Given a C++ program file, **Test2_Link.cpp** that implement a circular linked-list. You are provided with two classes, **Node** and **List**.

1. The **Node** class is fully provided (refer to **Figure 1**). This class represents a node in the circular linked-list. Each node contains a character and a link to the next node in the list.

```
1 class Node {  
2     char ch;  
3  
4     public:  
5         Node* next;  
6         Node(char c){ ch = c; next = NULL; }  
7         char getChar() { return ch; }  
8     };
```

Figure 1: Complete Node class definition

2. The **List** class is partially provided (refer to **Figure 2**). This class represents the circular linked-list itself.

```
1 class List {  
2     Node* head;  
3  
4     public:  
5         List() { head = NULL; }  
6         bool isEmpty() { return head == NULL; }  
7  
8         void insertNode(Node *);  
9         void deleteNode();  
10        string findNode(char);  
11        void displayList();  
12    };
```

Figure 2: Incomplete List class definition

You are required to complete several methods in this class based on the descriptions given below:

- a) **insertNode()** - This method should insert a new node at the **beginning** of the list. You need to consider two cases: (9.5 marks)
 - i) When the list is empty.
 - ii) When the list already has nodes.
- b) **deleteNode()** - This method should delete the node at the **end** of the list. You need to consider three cases: (12 marks)

- i) When the list is empty, display "*No nodes to delete.*" message.
 - ii) When the list has multiple nodes.
 - iii) When the list only has one node.
 - c) **findNode()** - This method should return the word "**True**" if the node being searched for is in the list, and return the word "**False**" if the node being searched for is not in the list. (8 marks)
 - d) **displayList()** - This method should display all nodes in the list. However, if the list is empty, it should display "*Nothing to display.*" message. (7.5 marks)
3. In the **main()** method. (13 marks)
- a) Create a list.
 - b) Using the list created in 3(a), you need to call the appropriate method(s) created in 2 to produce the output as shown in **Figure 3**.

```
No nodes to delete.  
A  
Z -> A  
Z is found in the list? True  
Z  
Nothing to display.
```

Figure 3: Output of node operations in the completed program

Question 2

[50 Marks]

Given a C++ program file, **Test2_Stack.cpp** with array implementation of **stack**. The implementation of the method `isEmpty()`, `isFull()`, `push()`, and `pop()` are given in the program. Based on the description below, complete the task of the implementation for `stackTop()`, `display()` and `checkSize()` methods. Then, using all of the defined methods, complete the `main()` function by appending the appropriate line of code at the space provided to generate the output shown in Figure 2(a), (b) and (c). You must **NOT change** any line of code regarding the definition of the given method.

Task 1 :

Complete the implementation of the `stackTop()` method to get the top element in the stack. (8 marks)

Task 2 :

Complete the implementation of the `display()` method to display all of the elements in the stack starting at the top. (12 marks)

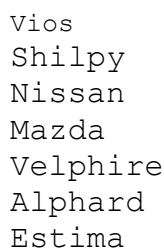
Task 3 :

Complete the implementation of the `checkSize()` method to get the number of elements in the stack. (8 marks)

Task 4 :

Complete the **main** method with a few calls and/or statements. Use the methods defined in the program to perform the following operations:

- Use the appropriate code that able to push `theCars` elements into the stack. Display all the elements in the stack as shown in Figure 2(a) (8 marks)



```
Vios
Shilpy
Nissan
Mazda
Velpshire
Alphard
Estima
```

Figure 2(a): After Task 4(i) is completed

- ii. Use the appropriate code that able to able to remove and display the removed 2 top elements in the stack as shown in Figure 2(b) (6 marks)

```
Remove Vios  
Remove Shilpy
```

Figure 2(b): After Task 4(ii) is completed

- iii. Display all the elements in the stack. Use the appropriate code that able to display the number of elements in the stack as shown in Figure 2(c) (8 marks)

```
Nissan  
Mazda  
Velpshire  
Alphard  
Estima  
  
#Cars in ShowRoom : 5
```

Figure 2(c): after Task 4(iii) is completed